PRODUCT · TECHNICAL · VISION DOSSIER

Patterns

Wildlife intelligence for spotted cats — turning camera-trap photos into a living database of named individuals and the population science behind them. No collars.

For scientists | For developers | For partners & investors

Version 2.0 · full dossier · **Launch species** Jaguar — ocelot & margay on roadmap
Status Build-ready specification · **Prepared** June 2026

CONTENTS

The whole picture, in one place

Five parts. Read end-to-end, or jump to your lens — vision, mechanics, the product visually, or the build.

PART I

Vision & product

What Patterns is, why it matters now, what it does, and why it is defensible. The orientation layer for anyone — a scientist, a partner, or an investor — meeting the product for the first time.

SECTION 01

Executive summary

Patterns is a wildlife-intelligence platform that identifies **individual** wild cats from camera-trap photos by their unique coat patterns, tracks their movement non-invasively, and turns the result into publishable population science.

What it is

A shared database of individually-identified spotted cats, fed by camera-trap photos from researchers and citizen scientists, with AI matching, a research-grade data pipeline, and an open-standards publishing path.

Who it's for

Field biologists and conservation NGOs as the core users; citizen scientists and reserves as contributors; funders, governments and the public as consumers of the resulting biodiversity data.

The wedge

Launch on **jaguars** — charismatic, threatened, heavily camera-trapped, and pattern-identifiable — then extend the same engine to ocelots, margays and other spotted felids.

The moat

A proprietary, growing dataset of confirmed individual identities that compounds with use (the data flywheel), plus open-standard interoperability that creates network effects rather than lock-out.

Why it wins

- **It produces the output scientists actually need** — detection-history matrices and density estimates — not just a photo gallery. That is the difference between a tool researchers *try* and one they *depend on*.
- **Every confirmed match makes the AI better.** The platform generates its own labelled training data, so accuracy and the dataset both compound — a moat that widens with adoption.
- **It is built on open biodiversity standards** (Camtrap DP, Darwin Core, GBIF), so data is publishable and poolable. That earns institutional trust and turns each new project into a network node.
- **Modern, mature UX** in a category dominated by aging research tooling — the experience itself is a wedge with the next generation of field scientists.

> **One rule defines the product's integrity:** verified researchers name the cats, and that name is the animal's canonical identity everywhere in the system. Identity is always human-confirmed and fully auditable — the AI proposes, people decide.

SECTION 02

The problem & the opportunity

Manual ID is the bottleneck

Camera traps produce millions of images; identifying which individual is in each one is slow, expert, manual work that rarely scales past one team.

Collars don't scale

GPS collars are invasive, costly, require capture and sedation, and only ever cover a handful of animals — useless for population-level monitoring.

Data dies in silos

Most studies end as spreadsheets and hard drives — unpublished, unstandardised, impossible to pool across regions or revisit.

Why now

AI re-ID has matured

Open foundation models for animal re-identification now make automated individual matching practical and accurate enough to put in front of scientists.

Camera traps are everywhere

Cheap, ubiquitous sensors mean the raw data already exists at scale — the constraint is turning it into knowledge, which is exactly the gap Patterns fills.

Biodiversity is being measured

Rising conservation funding and nature-related disclosure expectations are driving demand for credible, standardised population data.

> **The opportunity in one line:** the raw images, the AI, and the demand for biodiversity data all now exist — what's missing is the platform that connects them into trustworthy, publishable individual-level intelligence. That is Patterns.

Sustainability model · illustrative, not committed

Stream	Who pays	For what
Hosted research workspaces	Conservation NGOs, universities	Private projects, collaboration, analytics, priority processing.
Institutional / government licensing	Agencies, protected-area authorities	Regional monitoring programs and reporting dashboards.
Grants & conservation funding	Foundations	Core platform and open public layer, as conservation infrastructure.
Premium analytics & API	Research consortia	Density modelling, programmatic access, custom exports.

SECTION 03

The science — individual ID without collars

Spotted cats carry a natural fingerprint: no two individuals share the same rosette pattern. Patterns reads that fingerprint to recognise individuals across photos, locations and years.

This is an established method — **photographic capture–recapture**, or "capture–recapture without marks." Re-detections of the same individuals across many camera locations build a detection history for the whole population, which feeds spatial capture–recapture (SECR) models that estimate density, home range and population trend — the gold standard for monitoring elusive cats.

> **What it produces — and what it doesn't.** Not a live GPS dot. You get discrete re-detections at fixed camera locations, from which range and movement are reconstructed. For population science this is the more valuable dataset. "Tracking" here means reconstructing movement from re-sightings, not continuous telemetry — set that expectation early.

The species boundary, by design

Pattern-identifiable ✓

Jaguar, ocelot, margay, oncilla, leopard, cheetah, serval — unique coat markings act as an individual fingerprint.

Not pattern-identifiable ✕

Puma, jaguarundi and other uniform-coat cats can't be individually identified from pattern. "Supports individual ID" is an explicit per-species flag, never a global assumption.

SECTION 04

Feature catalogue

The full product surface, grouped by capability area.

Capture & identification

- Single & bulk upload from camera-trap SD cards
- Auto location via EXIF → camera station → manual pin
- Blank-frame filtering & animal cropping
- Automatic species classification
- Coat-pattern individual matching with ranked candidates
- Per-side (left/right) identity handling

Identity & naming

- Researcher naming with admin approval
- Canonical names used app-wide
- Human-in-the-loop match confirmation
- Merge / split with full audit trail
- Individual profiles with timeline & gallery
- Follow an individual

Community & data quality

- Observations feed (iNaturalist-style)
- Quality grades up to Research Grade
- Multi-researcher agreement
- Annotations (sex, age, condition, behaviour, scars)
- Contributor leaderboards & engagement

Research & analytics

- Filterable research data table
- Detection-history (capture) matrix
- SECR density & abundance
- Range, movement & co-occurrence metrics
- Sampling-effort (trap-night) tracking
- Meaningful scientist alerts

Collaboration & publishing

- Projects as the unit of work & publication
- Camera-station registry
- Export to CSV, Camtrap DP, Darwin Core
- GBIF publishing with citable DOI
- Research API

Admin & governance

- Name-approval, review & merge queues
- Researcher verification
- Role-based access & coordinate generalisation
- Master map & cross-project oversight
- Full audit logging

SECTION 05

Landscape & moat

Strong tools exist for parts of this problem. None integrates individual ID, a modern community layer, and research-grade population output in one place.

Capability	Patterns	Wildbook	iNaturalist	Wildlife Insights
Individual (not just species) ID	●	●	—	—
Community observation feed	●	limited	●	limited
Research-grade gating	●	partial	●	partial
Density / SECR-ready output	●	via export	—	occupancy
Open standards & GBIF	●	●	●	●
Modern, integrated UX	●	dated	●	enterprise

Comparison reflects general positioning of capable, actively-developed platforms; each has strengths Patterns should interoperate with rather than dismiss.

The moat, concretely

Data flywheel

Confirmed matches become labelled training pairs. The model and the dataset both compound with usage — a self-reinforcing accuracy advantage competitors can't shortcut.

Switching cost

Once a project's named individuals, history and analytics live in Patterns, that identity record is the project's institutional memory. Leaving means abandoning it.

Network effects

Standards-based interop means each new project and region makes pooled data more valuable for everyone — growth compounds reach rather than fragmenting it.

PART II

How it works

The mechanics, end to end — the data model and the naming rule, who can do what, how people get in, the journeys they take, and the recognition engine that powers it all.

SECTION 06

Core concepts, the data model & naming

One decision underpins everything: separate the **animal** from the **sighting**.

An **Individual** is one animal — one profile, one name, named exactly once, with a pattern signature and status. A **Detection** is a single photo/video: media, location, time, uploader, and a link to the Individual it matched. Many detections attach to one Individual. That separation makes "named once," "notify on re-detection," and "no duplicate names" fall out naturally.

Aligned to an open standard from day one

Patterns entity	Camtrap DP equivalent	Holds
Camera Station	Deployment	Fixed trap location, active dates, camera model, sampling effort.
Detection (media)	Media	The file, capture timestamp, EXIF, contributor.
Detection (observation)	Observation	Species, matched Individual, confidence, body side, reviewer status.
Individual	Individuals resource	Animal: stable ID, species, name, naming claim, pattern embeddings, status.

Researchers name the cats — and that name is the cat's name everywhere

The single naming rule the whole platform is built around. When a **verified researcher** names a new individual and an admin approves it, that name becomes the **canonical label across the entire app**: profile, every map pin, the observations feed, every notification, all exports, the API, and any citation. One name, one namer, credited permanently.

Before naming

Referred to everywhere by a stable system ID — e.g. `JAG-0247` — so it can still be tracked while unnamed.

After naming

The researcher's chosen name **replaces that ID app-wide** the instant it's approved. Contributors and viewers see and use it but can never create or change it.

Naming is a researcher privilege and an act of record. The name a researcher gives is what every user, export, and published dataset will call that cat — so it carries scientific weight and passes through admin approval to prevent duplicates.

Identity is a claim, not a fact. Because identity is decided from the pattern, every assignment is auditable and reversible. Merges and splits never delete data; they re-point detections and keep history. The earliest approved name wins a merge.

SECTION 07

Roles & permissions

Admin

Coordinator

Referees naming, merges, review; manages users & verifies researchers. Sees everything.

Researcher

Verified scientist

Names individuals (canonical, admin-approved), confirms matches, tracks cats, exact coords on own data.

Contributor

Field / citizen

Uploads & sees matches. No naming or identity judgement. Own-upload coords only.

Viewer

Public

Browses profiles & map, read-only. Locations always generalised.

Capability	Admin	Researcher	Contributor	Viewer
Upload photos / videos	●	●	●	—
Name a new individual	●	●	—	—
Approve / reject names	●	—	—	—
Confirm / reject AI matches	●	●	—	—
Merge / split profiles	●	request	—	—
Re-detection notifications	●	●	own	—
See exact coordinates	all	own data	own uploads	fuzzed
Manage users & verify researchers	●	—	—	—
Export / publish (GBIF)	●	own project	—	—

● allowed **amber** scoped / conditional — not allowed

SECTION 08

Authentication & sign-in

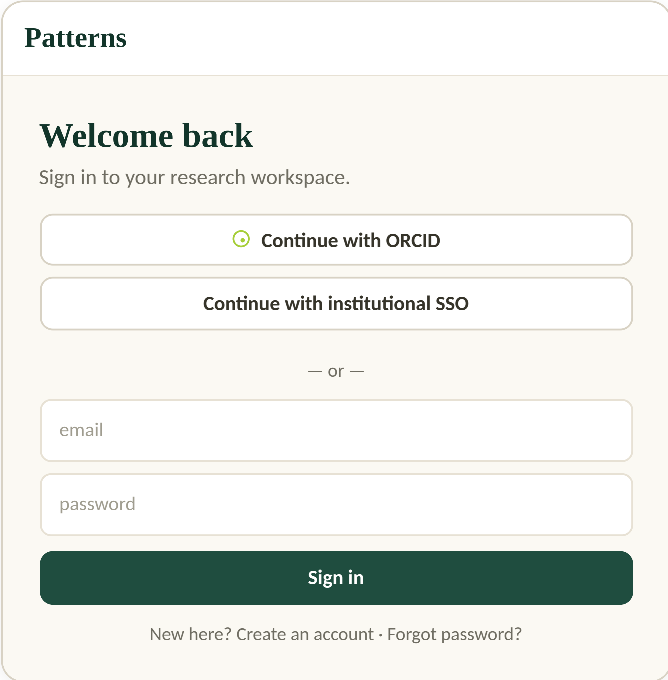
Getting in is fast for everyone, but scientific authority is earned. Sign-in is built around researcher identity from the start.

Ways to sign in

- **ORCID** — the global researcher identity standard. One click associates an account with a verified scholarly identity, and is the fast path to researcher verification.
- **Institutional SSO** (Google / Microsoft / SAML) — for universities and NGOs that manage accounts centrally.
- **Email + password** — for citizen contributors and anyone without the above, with verification by email link.

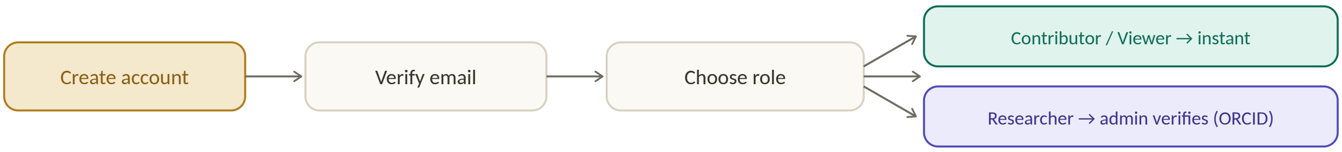
Account security

- Email verification required before first upload.
- Optional MFA; **required for Admins**.
- Sessions are short-lived JWTs carrying the user's role & verification state; refresh tokens rotate.
- Password reset by signed, expiring email link.



Sign-in — researcher identity (ORCID / SSO) is the primary path.

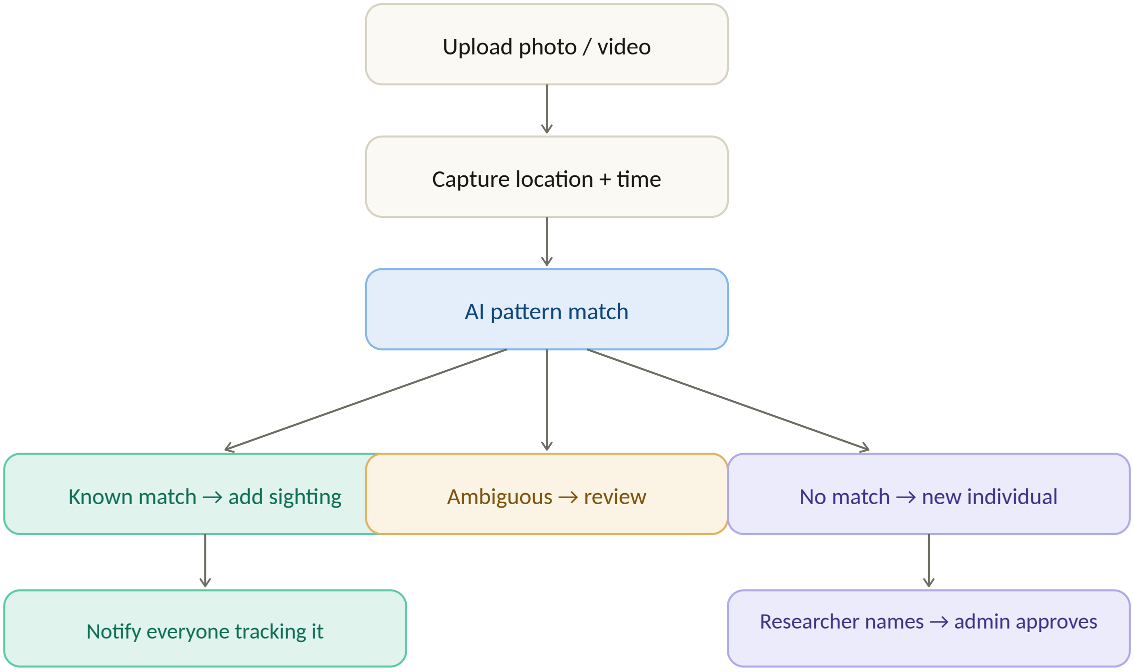
Sign-up → role → verification



Contributors and viewers are live immediately; researcher rights unlock after verification.

> **Verification gates authority, not access.** A pending researcher can upload and browse from minute one — they just can't name individuals or confirm identities until an admin verifies their affiliation (ORCID makes this near-instant).

End-to-end user journeys



The core detection-and-naming journey — three outcomes from one AI match.

Upload & detection

1. **Upload** one or many files (batch from an SD card).
2. **Location & time** resolve via EXIF → station → manual pin.
3. **Detect, classify, match** run automatically; a ranked candidate list returns with scores.
4. **Detection is pinned** and tagged known / review / new, scoped to the viewer's role.

Re-detection

1. **New upload matches** a known cat with high confidence.
2. **Sighting appends** to the profile; movement line redraws.
3. **Trackers are notified** — "Itzel re-detected 40 km north."

Naming a new individual

1. **No confident match** → new Individual gets a temp ID, marked unnamed.
2. **Researcher confirms** it's new and proposes a name (uniqueness checked).
3. **Admin queue** shows photos, proposer, near-matches.
4. **Approve / reject / merge** — on approve the name locks and goes canonical app-wide.

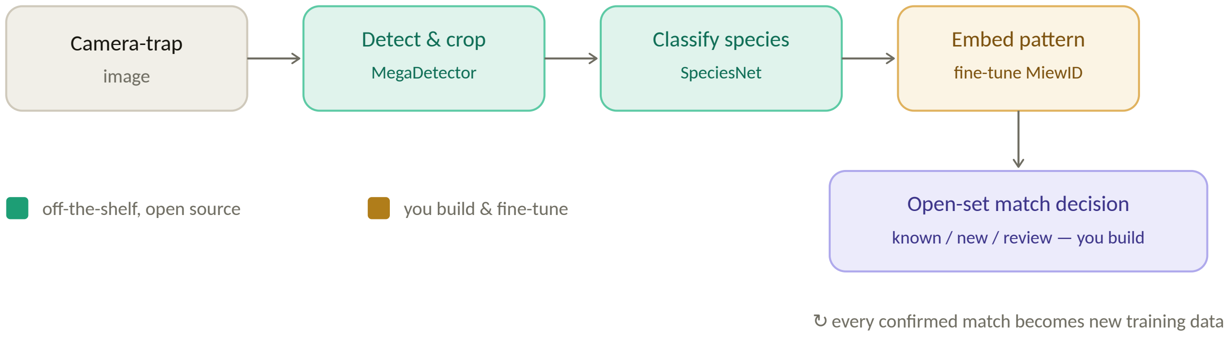
Review & merge

1. **Mid-confidence** matches wait in a human-review queue — never auto-decided.
2. **Duplicates merge** under admin; earliest approved name wins.
3. **Nothing is deleted** — merges are logged and reversible.

SECTION 10

The recognition engine

"From scratch" does not mean "from zero." Three of four pipeline stages are solved by free, open tooling; engineering concentrates on the fourth.



The four-stage pipeline. Effort concentrates on the embedding and the open-set decision.

Stages 1–2 · Detect & classify

MegaDetector finds animals and filters blanks; **SpeciesNet** labels species on its output across ~2,000 species. Both free and open — do not rebuild; fine-tune species if needed.

Stage 3 · Embed the pattern

Deep metric learning maps each crop to an embedding — the pattern signature. Start from **MiewID**, index with **pgvector / FAISS**, match by nearest neighbour per side.

The open-set problem is why human review is mandatory. Deciding "new vs already-in-the-database" is hard even for strong models. The AI proposes a ranked candidate list; a human confirms. It never auto-creates or auto-merges an individual — exactly how mature platforms operate, and what keeps the science defensible.

The data flywheel

Every confirmed match is a labelled pair; every rejection a hard negative. Feed them back to fine-tune. The platform makes its own training data — the reason to own the engine.

Field realities

- Grayscale IR night shots — augment for it.
- Blur & partial bodies — degradation augmentation.
- Left vs right is a hard dimension.
- Cold start: zero-shot embeddings + public datasets.

SECTION 11

The observation & community layer

Patterns borrows what works from iNaturalist — an open, community-driven observation feed — and adds what it lacks: **individual-level identity**.

Data quality grades — what counts as science

Grade	Meaning	Counts toward science?
Casual	Low quality, no animal found, or missing location/date.	No
Needs ID	AI matched or flagged, awaiting human confirmation (the review queue).	Not yet
Confirmed	A researcher confirmed the individual match.	**Provisionally**
Research Grade	Species + individual + valid location + valid date. Multi-researcher agreement raises it further.	**Yes**

Agreement
Multiple researchers confirm; disagreement re-opens review.

Annotations
Sex, age, condition, behaviour, scars — corroborate ID.

Projects
Group stations, observations & members; the unit of publication.

Species & individual pages
Aggregate range & count; follow an individual.

The observations feed

Naming in action: named cats (**Itzel**, **Balam**) appear by their researcher-given name everywhere; the unnamed one shows its system ID (**JAG-0247**) until a researcher names it.

> **Engagement that serves the science.** Leaderboards, "your photo helped confirm Itzel," and follow-an-individual alerts keep contributors active — and active contributors mean more detections, which means better population data.

SECTION 12

Researcher data & research-grade output

Where Patterns stops being a gallery and becomes population science — turning confirmed observations into the exact inputs researchers need.

12.1 · The data view

Patterns Search… **Export** AR

- Overview
- Data
- Analytics
- Individuals
- Map

Osa Jaguar Project

Data

Species: Jaguar | Grade: Research ▾ | 2023–26 | Station: all | Side: any

INDIVIDUAL	STATION	DATE	SIDE	SEX	CONF.	BY
Itzel	OSA-04	2026-05-02	L	F	0.97	M. Torres
Balam	OSA-11	2026-04-28	R	M	0.99	J. Núñez
Itzel	OSA-07	2026-04-19	L	F	0.95	field cam
Kanil	OSA-04	2026-04-11	R	M	0.96	A. Reyes

Every row is provenance-stamped — who, where, when, side, confidence, grade. Filters save; exports honour role-based coordinate generalisation.

12.2 · The detection-history matrix · the feature that matters most

The single most valuable output: a capture–recapture matrix of **individuals × sampling occasions**. This is the direct input to SECR and mark–recapture density models. No photo app produces it — it is the reason a serious researcher adopts Patterns.

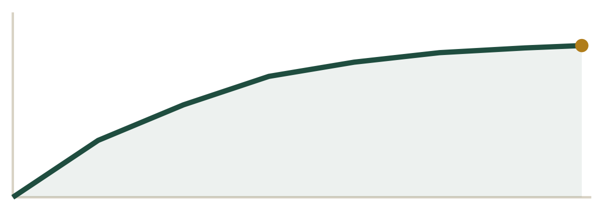
INDIVIDUAL	OCC 1	OCC 2	OCC 3	OCC 4	OCC 5	OCC 6
Itzel	●	·	●	●	·	●
Balam	·	●	·	·	●	·
Kanil	●	●	·	●	·	·

● detected · not detected. Export as a standard encounter-history file for SECR / Program MARK.

12.3 · Population analytics

312	1.8	61%	4,210
INDIVIDUALS	DENSITY /100KM²	RECAPTURE RATE	TRAP-NIGHTS

INDIVIDUAL ACCUMULATION



New individuals per survey month — flattening signals near-complete coverage.

- SECR density & abundance with confidence intervals.
- Per-individual range (MCP), max displacement, stations visited.
- Diel activity patterns & station co-occurrence.
- Sampling effort auto-computed from station active dates.

12.4 · Alerts, export & API

Meaningful alerts

"An individual you study reappeared after 8 months" · "a possible new individual is in your study area" · survival & movement signals, not noise.

Export

CSV · Camtrap DP · Darwin Core · encounter-history files for SECR. Coordinate generalisation enforced per role.

Research API

Programmatic pull of research-grade data with a citable DOI on publication.

Why this makes Patterns really helpful: a researcher's slowest work is going from thousands of photos to a defensible density estimate. Patterns collapses it: confirmed sightings → research-grade observations → detection-history matrix + effort → density. The photos are the means; the population estimate is the payoff.

PART III

The product, visually

The brand and design system that make Patterns feel mature and trustworthy, the key screens in high fidelity, and the map system that anchors the whole experience.

SECTION 13

Design system & brand

A calm, scientific, field-grade aesthetic — deep forest greens, a warm rosette gold, generous space, and the rosette motif as the brand's signature.

Palette

Forest · #12352A
primary, surfaces

Rosette gold · #B07D1A
accent, action

Teal · #0F6E56
confirmed, success

Canvas · #FBF9F3
warm paper

Plum · new individual

Sky · informational

Red · alerts

Ink · #17160F text

Typography

Patterns
Charter — display & headings

The quick brown jaguar — body & UI text
Carlito — interface, body, tables

`JAG-0247 · 0.97 conf · mono for IDs & data`

Components

Primary Action Ghost

Admin Researcher Research Grade Needs ID

filter chip Species: Jaguar

All Mine Following

Iconography

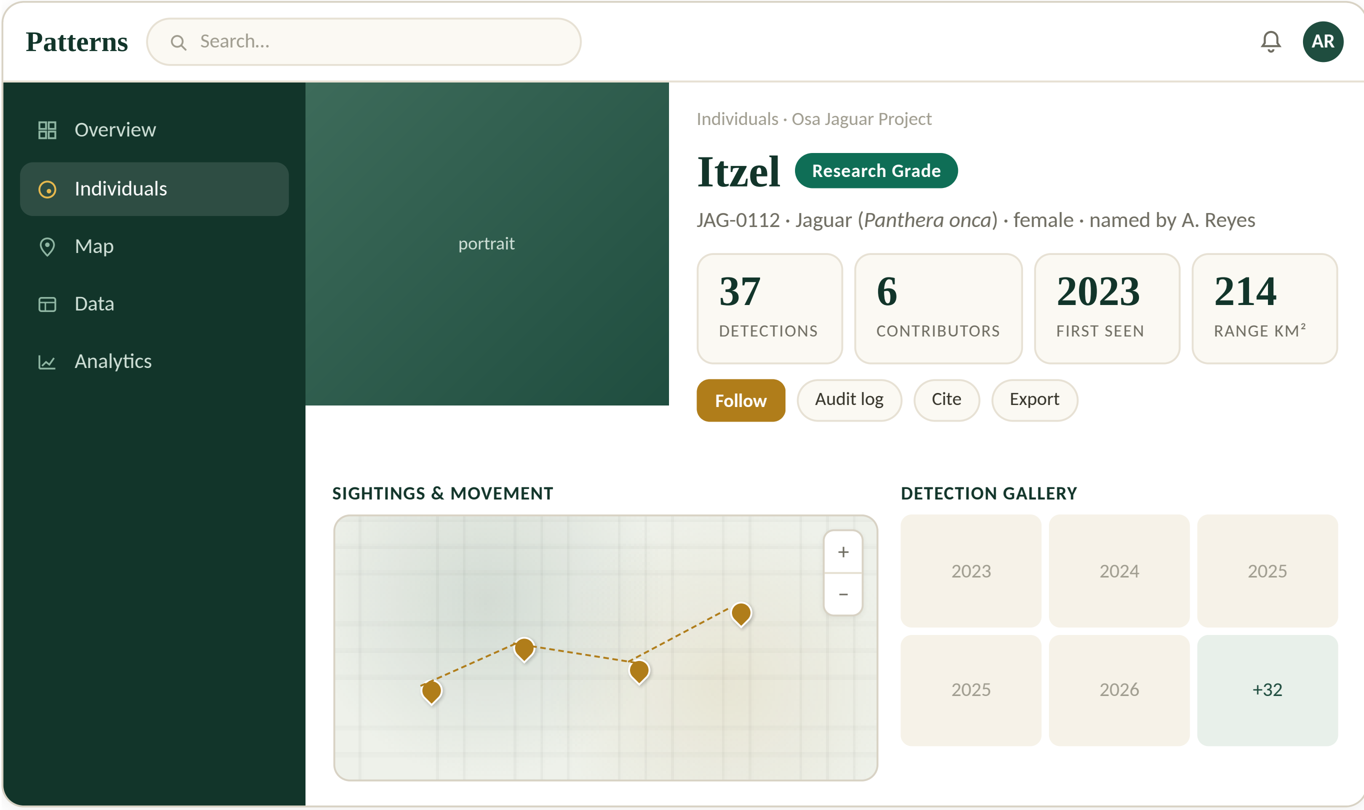
individuals map data analytics stations upload people projects

The rosette — a ring with an inner spot — is both the app's identity icon and a literal nod to the coat patterns the whole platform reads.

SECTION 14

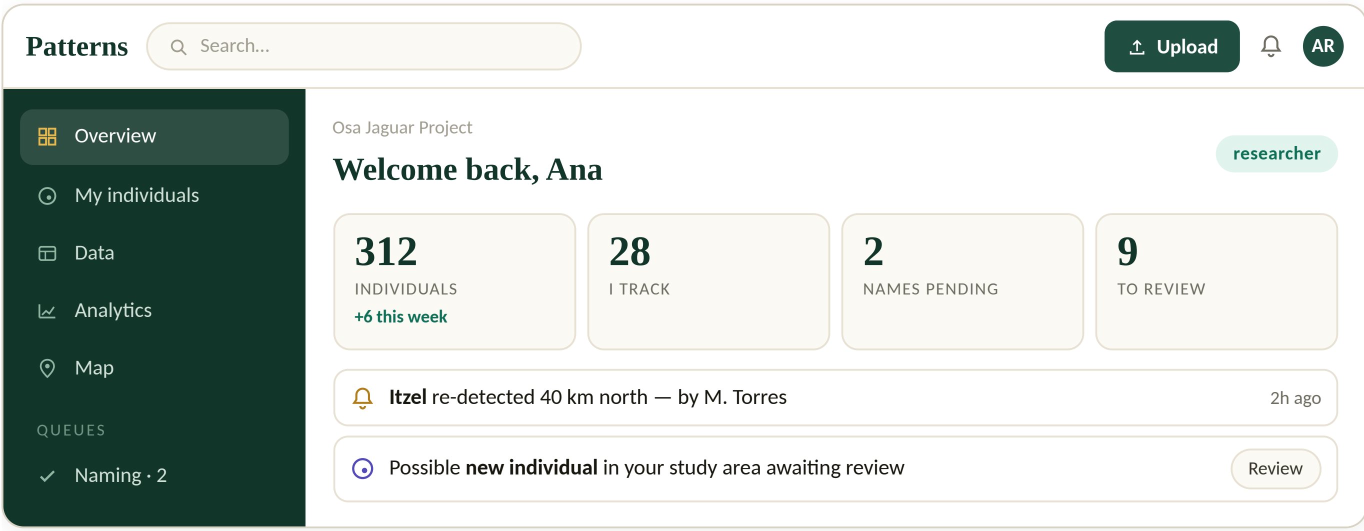
Screens — high-fidelity UI

14.1 · Individual profile — the showpiece

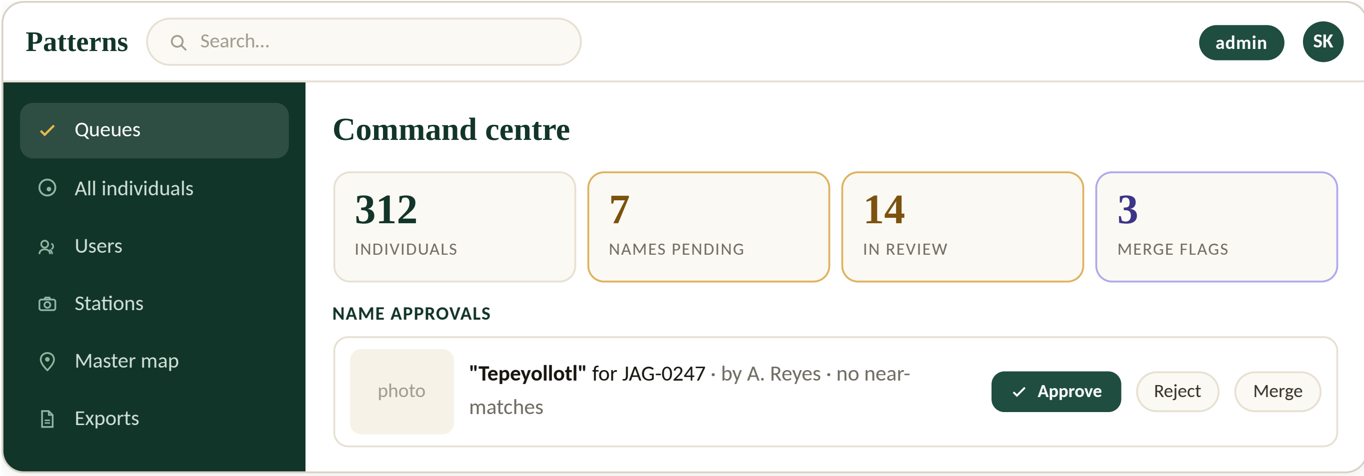


The page researchers screenshot and cite — hero portrait, canonical name & grade, stats, movement map, gallery, and one-tap follow / cite / export.

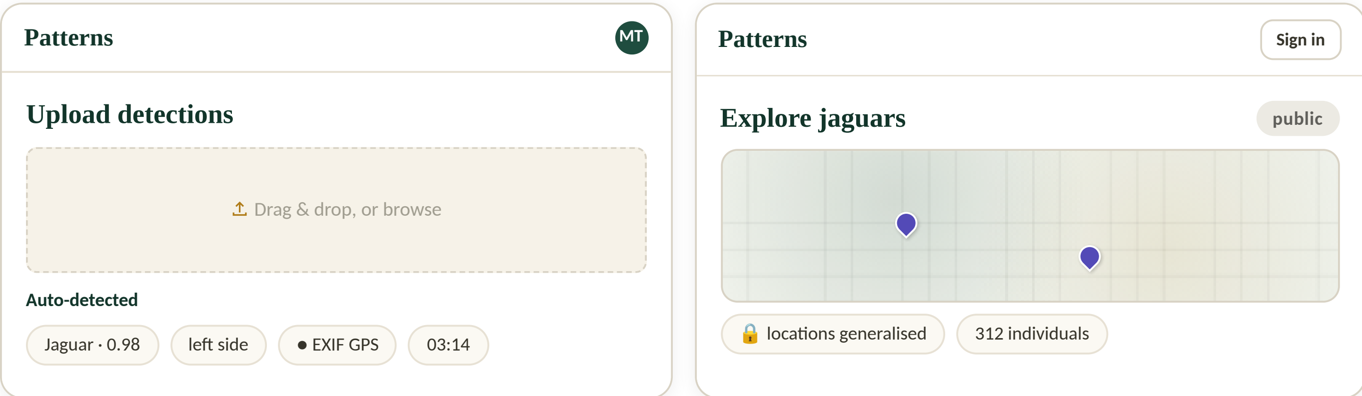
14.2 · Researcher dashboard



14.3 · Admin dashboard — the command centre



14.4 · Upload & 14.5 · Public explore



Upload runs detect→species→match live, auto-filling location. **Public explore** shows generalised pins only — no exact location of a threatened animal ever reaches an unauthenticated viewer.

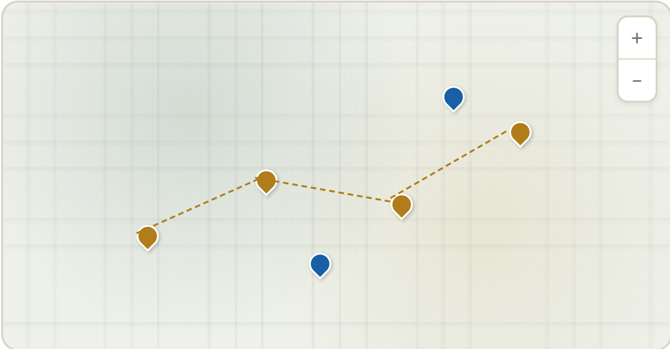
SECTION 15

The map system

Maps are the heart of the product — but for an endangered animal, a precise location is a poaching map. Sensitivity is built in, not bolted on.

Four map views

- **Individual sightings map** — one cat's detections with dashed movement lines in date order.
- **Station map** — registered camera stations, active periods, sampling effort.
- **Population / density map** — aggregated detections as an SECR density surface (researchers & admins).
- **Public map** — generalised pins only.



this individual   camera station   generalised (public)

Movement reconstructed from re-detections — not a continuous track.

Location capture — the EXIF → station → pin ladder

Most camera traps don't embed GPS, so phone-photo metadata alone yields a sparse, inaccurate map. Each detection resolves location in priority order: **(1)** EXIF GPS, **(2)** the registered **camera station**'s fixed coordinates, **(3)** a manual pin. Stations are also the basis for sampling-effort calculations.

Coordinate generalisation by role

Role	Resolution shown	Rationale
Admin	Exact, all data	Needs full precision to resolve conflicts and merges.
Researcher	Exact for own / collaborating projects; generalised elsewhere	Science needs precision; other teams' sites stay protected.
Contributor	Exact for own uploads; generalised elsewhere	Verify own contribution without exposing the network.
Viewer	Grid cell / region only	Public access never reveals an exact threatened-animal location.

Standard, not optional. The Camtrap DP standard explicitly supports restricting threatened-species imagery and generalising coordinates. Store exact coordinates once, derive the generalised version, serve by role — never the reverse.

PART IV

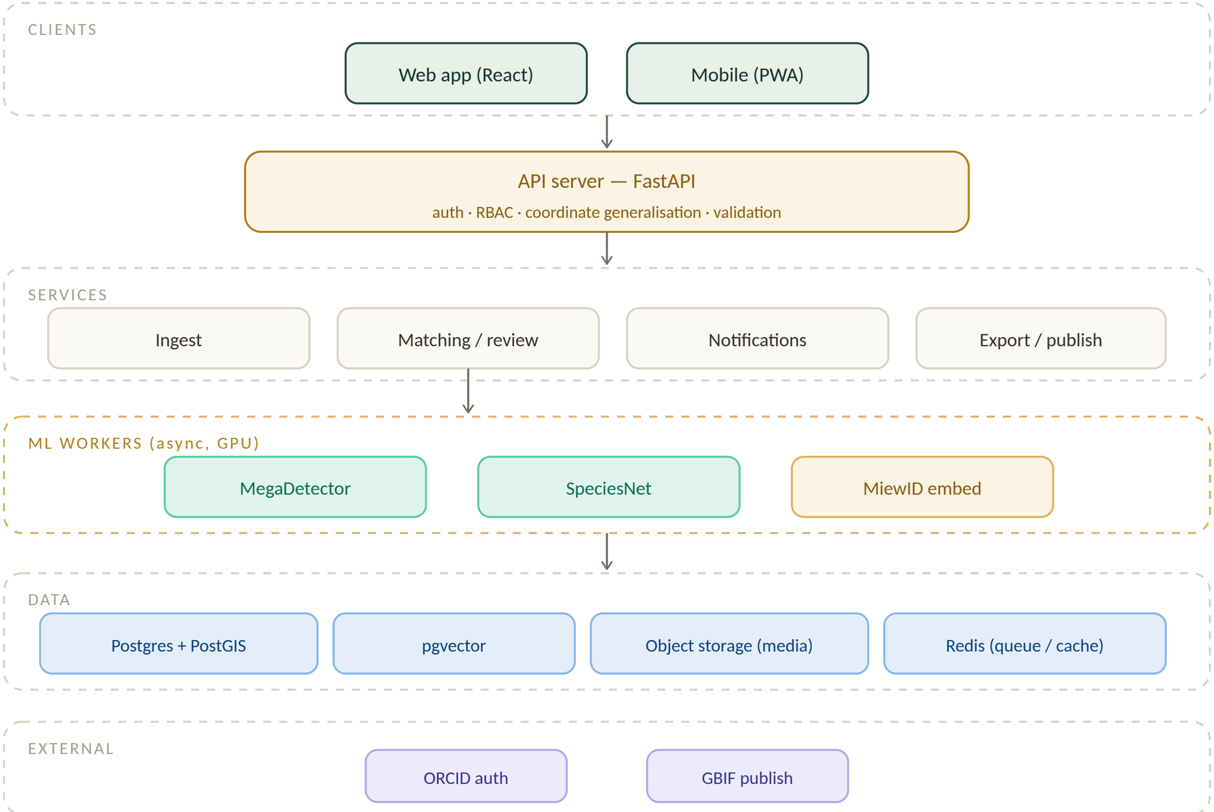
Build it

Everything an engineer needs to start: the system architecture, the recommended stack, the data schema, the API surface, infrastructure and security, and a phased roadmap.

SECTION 16

System architecture

A conventional, boring-in-the-good-way architecture: a typed web client, an API that owns access control, async ML workers, and a spatial database with a vector index alongside it.



Clients talk only to the API; the API enforces access control and generalisation; heavy ML runs async; data splits across a spatial DB, a vector index, and object storage.

> **Design principle:** the API is the only place that knows a user's role, so coordinate generalisation and permissions are enforced server-side on every response — never trusted to the client.

SECTION 17

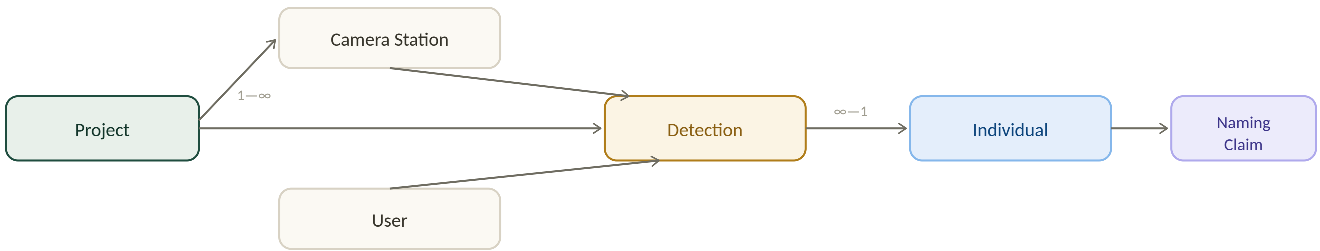
Tech stack

Layer	Choice	Why
Frontend	React + Next.js, TypeScript, Tailwind	Mature, typed, fast to build; SSR for public pages & SEO on profiles.
Maps	MapLibre GL + vector tiles	Open, no per-view licensing; custom styling and clustering for stations & sightings.
Backend / API	Python, FastAPI	Same language as the ML stack; async, typed, OpenAPI out of the box.
Database	PostgreSQL + PostGIS + pgvector	One database for relational data, geospatial queries, *and* embedding nearest-neighbour search.
Object storage	S3-compatible	Cheap, scalable media storage with signed URLs; originals never served directly.
Queue / workers	Redis + Celery / RQ	Async ML pipeline; uploads return immediately, processing happens in the background.
ML	PyTorch · MegaDetector · SpeciesNet · MiewID	Open foundation models; only the matching head is fine-tuned in-house.
Auth	OAuth/OIDC (ORCID, Google, Microsoft), JWT	Researcher-first identity; institutional SSO; stateless sessions carrying role + verification.
Infra	Docker, autoscaling cloud, GPU pool for inference	CPU API tier scales separately from the GPU ML tier to control cost.

SECTION 18

Data schema

Camtrap-DP-aligned core entities. Relationships first, then the key fields.



A project owns stations, detections and individuals · users upload detections · many detections resolve to one individual · one individual has one approved naming claim.

individuals

```
id · project_id · species · sex · status
(unnamed/named/under_review) · name
· name_claim_id · created_at
```

detections

```
id · project_id · station_id · media_id
· individual_id · species · side(L/R)
· captured_at · confidence · grade
· reviewer_id · uploader_id
```

media

```
id · storage_key · exif_gps · captured_at
· embedding (vector, pgvector) · width/height
```

camera_stations

```
id · project_id · name · location (PostGIS)
· active_from · active_to · camera_model
```

naming_claims

```
id · individual_id · proposed_name
· proposer_id · status(pending/approved/
rejected) · approver_id · decided_at
```

users · projects · audit_log

```
users: id · role · verified · orcid
projects: id · name · members[]
audit_log: actor · action · entity · diff · ts
```

Embeddings live beside the data: the pattern vector is a `pgvector` column, so a match is a single nearest-neighbour SQL query filtered by species and side — no separate vector service to operate at this scale.

SECTION 19

API specification

REST, JSON, OpenAPI-documented. Every response is filtered by the caller's role server-side.

Method & path	Purpose
`POST /auth/orcid`	Begin ORCID OAuth; returns session JWT with role & verification state.
`POST /uploads`	Upload media (signed URL); enqueues detect→species→match. Returns a job id.
`GET /detections/{id}`	Detection with ranked candidate individuals & scores (coords generalised by role).
`POST /detections/{id}/confirm`	Researcher confirms or rejects a match (human-in-the-loop).
`POST /individuals/{id}/name`	Researcher proposes a name → opens a naming claim.
`POST /naming-claims/{id}/decision`	Admin approves/rejects; on approve the name goes canonical.
`POST /individuals/merge`	Admin merges duplicates; logged & reversible.
`GET /projects/{id}/data`	Filterable detection/individual data for the research table.
`GET /projects/{id}/capture-matrix`	Detection-history matrix & encounter-history export for SECR.
`POST /projects/{id}/export`	Generate CSV / Camtrap DP / Darwin Core package.

Sample match response


```
{ "detection_id": "d_8821", "species": "Panthera onca",
  "side": "L", "grade": "needs_id",
  "candidates": [
    { "individual": "Itzel", "id": "JAG-0112", "score": 0.94 },
    { "individual": "JAG-0231", "score": 0.61 } ],
  "decision": "awaiting_review" }
```

SECTION 20

Infrastructure, security & deployment

Deployment

- Containerised services; CPU API tier and GPU ML tier scale independently.
- Async workers process uploads so the UI never blocks.
- Bulk-import path for dumping whole SD cards (thousands of images).

Security & privacy

- Role-based access enforced at the API; coordinate generalisation applied to every response.
- Signed, expiring URLs for media; originals never public.
- Threatened-species locations sensitive by default.
- Full audit log on every identity decision.

Reliability

- Daily encrypted backups of DB + object storage; point-in-time recovery.
- Embeddings re-computable from media, so the vector index is rebuildable.
- Idempotent ingest — re-processing a file never duplicates a detection.

Observability

- Track review-queue size & time-to-decision as product SLAs.
- Monitor match-confidence distribution & open-set accuracy over time.
- Per-project processing dashboards.

SECTION 21

Build roadmap

Phase	Goal	What ships
0 · Foundations	Schema & access	Camtrap-DP data model, ORCID/SSO auth & the four roles, station registry, coordinate generalisation.
1 · Matching loop	Working ID	MegaDetector + SpeciesNet front end, MiewID embeddings, pgvector lookup, human-verification screen — upload → candidates → confirm.
2 · Profiles & community	The product	Individual profiles, sightings maps, researcher naming & approval, observations feed with quality grades, notifications, dashboards.
3 · Flywheel	Owned engine	Fine-tune the matching head on confirmed matches; tune confidence bands; measure open-set accuracy.
4 · Open science	Publish & expand	Research data view, detection-history matrix, population analytics & API, GBIF export, public explore, second species (ocelot).

Open decision: whether to fine-tune your own matching head (phase 3) or lean on open foundation models longer — a cost/control trade-off worth revisiting once phase 1 shows real accuracy on your own jaguar data.

PART V

Appendix

Open standards and the publishing pipeline that make Patterns real science infrastructure — plus a glossary of the terms used throughout this dossier.

SECTION 22

Standards, publishing & glossary

Because the schema follows Camtrap DP, a project's data exports as a valid package, converts to Darwin Core, and publishes to GBIF — citable and poolable.

Patterns DB → Camtrap DP export → Darwin Core → GBIF

Glossary

Term	Meaning
Capture–recapture	Estimating a population from repeat detections of identifiable individuals.
SECR	Spatial explicit capture–recapture — models that estimate density from where and when individuals are detected.
Detection-history matrix	Individuals × sampling occasions grid of detections — the input to SECR.
Open-set recognition	Deciding whether an individual is known *or* entirely new — the hard ML problem at the core.
Embedding	The numeric pattern signature of a coat, used for nearest-neighbour matching.
Research Grade	An observation with confirmed species, individual, location and date — eligible for science.
Camtrap DP / Darwin Core / GBIF	The open data standard, the biodiversity format, and the global database Patterns publishes to.
MegaDetector / SpeciesNet / MiewID	Open models for animal detection, species classification, and individual re-identification.

Patterns — Product, Technical & Vision Dossier · v2.0 · prepared June 2026. External open projects (MegaDetector, SpeciesNet, MiewID, Camtrap DP, GBIF, ORCID) are referenced for integration; verify current versions and licences. Mockup names and figures are illustrative placeholders.